Phonological Assembly Circuits in GPT-2

How Transformers Compute Sound Structure from Orthography

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Overview

Part 1: The Question

Why phonological circuits?

The MI literature has characterized circuits for:

- Syntax: IOI (Wang et al., 2023), subject–verb agreement, reflexive anaphora
- Retrieval: induction heads, greater-than
- Factual: capital cities, birth years

Gap: zero circuit-level work on phonological computation.

Yet LLMs demonstrably encode sound structure:

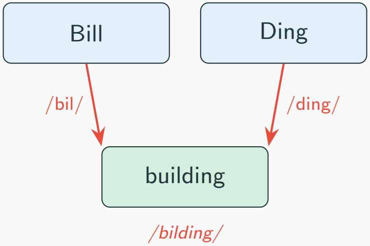
- Linear probes on GPT-2 mid-layers: 63–79% rhyme detection (Liao & Shi, 2026)
- Above-chance grapheme-to-phoneme conversion (PhonologyBench, 2024)

Question: *Which attention heads compute phonological knowledge?*

The puzzle

BPE tokenization encodes **orthography**, not phonology.

Yet the model can do this:



Cross-boundary phoneme fusion from purely orthographic input.

How? Which heads? Is there a dedicated circuit?

Six phonological operations

Op	Linguistic term	Example	Computation	N
1	Rhyming hypocorism	Richard → Dick	Irregular stored lookup	32
2	Clipping (apocope)	Timothy → Tim	First-syllable truncation	90
3	Initialism	J → Jay	Grapheme → phoneme name	30
4	**Oronym**	**Bill Ding → building**	**Cross-boundary fusion**	**99**
5	Homophone	Neil → kneel	Phoneme-identity detection	120
6	Folk etymology	microwave → Michael Wave	Reverse decomposition	75

Each requires a **linguistically distinct** computation.

→ Predicts **distinct circuit architectures**.

Operations as a computational taxonomy

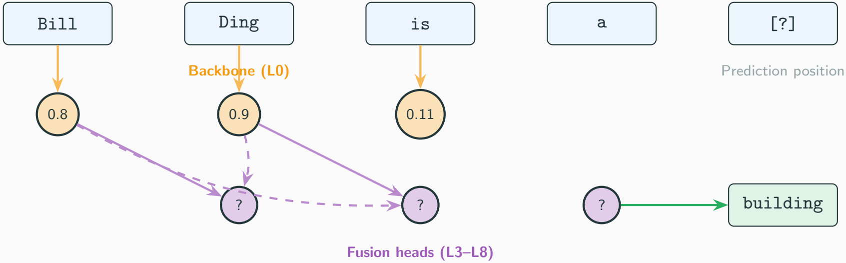
Stored lookup	Regular rule	Compositional
Op 1 Hypocorism	Op 2 Clipping	Op 4 Oronym
Richard → Dick	*Timothy → Tim*	*Bill Ding → building*
Op 5 Homophone	Op 3 Initialism	Op 6 Folk etymology
Neil → kneel	*J → Jay*	*microwave → Michael Wave*

Increasing computational complexity →

Stored → Regular → Compositional: each category demands more computation.

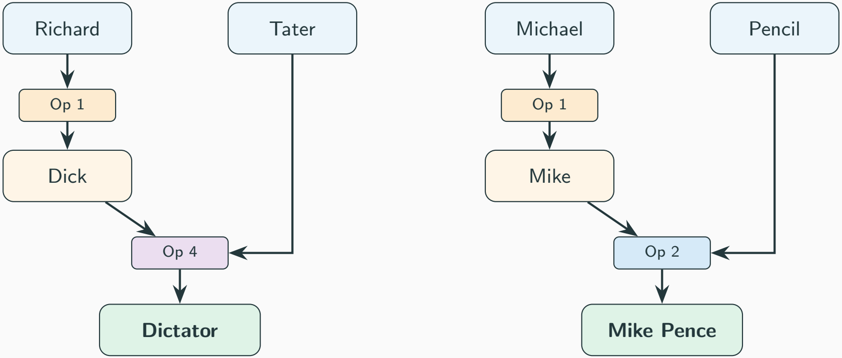
Op 4 (oronym) requires **cross-token phoneme assembly** — the hardest operation.

The oronym problem



Expected: fusion heads attend *across* the word boundary, composing phonemes from both "Bill" and "Ding" into "building."

Chained operations: composing circuits



Op 1 + Op 4　　　　*Op 1 + Op 2*

Compositionality test: ablate Op 1 circuit $\Rightarrow$ compound breaks, Op 4 survives.

Ablate Op 4 circuit $\Rightarrow$ compound breaks, Op 1 survives.

Part 2: Methods

Minimal-pair datasets

Each pair follows [BLiMP](#) format (Warstadt et al., 2020): {`good`, `bad`, `target_token`, `foil_token`}

Operation	Good prompt	Target / Foil
Op 1	"People call Richard by the name"	Dick / Mark
Op 2	"Timothy goes by"	Tim / Tom
Op 3	"The letter J is pronounced"	Jay / Gee
Op 4	"Bill Ding sounds like"	building / nothing
Op 5	"Neil sounds like"	kneel / wheel
Op 6	"Mike Rowave sounds like"	microwave / nothing

Evaluation: logit difference at final position.

Gate: >70% accuracy before circuit discovery.

Ops 2, 3, 5 are procedurally expandable (CMU Pronouncing Dictionary).
Ops 1, 4, 6 are controlled vocabularies.

The dataset: selected examples

Input	Target	Operation
Richard Tater	Dictator	Op 1 + Op 4
William Ding	Building	Op 1 + Op 4
Elizabeth Aardvark	Lizard	Op 2 + Op 4
Bill Ding	Building	Op 4 only
Justin Time	Just in time	Op 4 only
Brock Lee	Broccoli	Op 4 only
Barb Dwyer	Barbed wire	Op 4 only
Mike Rowave	Microwave	Op 6 (reverse)
Dan Druff	Dandruff	Op 4 only
Herb Ivore	Herbivore	Op 4 only

Tokenization audit

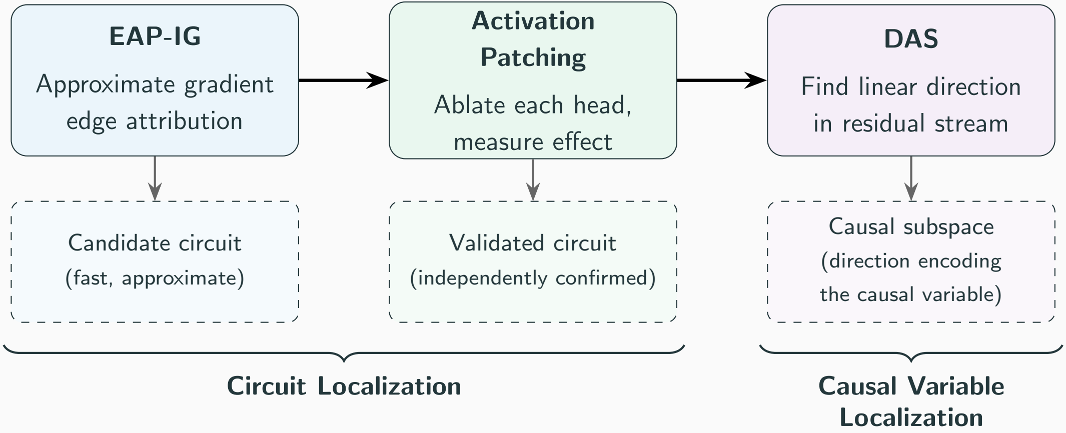
BPE tokenization can break phonological tasks:

Good:	`Bill`	`Ding`	✓ single tokens
Bad:	`build`	`ing`	× split target
Slash:	`B` / `i` / `l` / `l`		character-level input

Following Liao & Shi (2026): if slash-delimited input improves Op 4 accuracy, the circuit *exists* but is bottlenecked by tokenization.

$\Rightarrow$ Distinguishes tokenization bottleneck from missing circuit.

Three-stage analysis pipeline



Part 3: Hypotheses

Prior circuits converge on the same layers

Known GPT-2 circuits reuse the **same layers** across unrelated tasks:

Task	Key heads (layers)	Source
IOI	S-inhibition (L7–8), name movers (L9–10)	Wang et al. 2023
Greater-than	Attn (L5–9), MLPs (L8–11)	Hanna et al. 2023
Induction	Prev-token (L2, L4), induction (L5–7)	Olsson et al. 2022
SVA	Top heads span L0, L6, L9–11	Lazo et al. 2025

L11: Readout / unembedding

L7–10: Convergence zone (IOI, GT, SVA, induction all active here)

L5–7: Intermediate (induction, greater-than attention)

L0–4: Early processing (prev-token, duplicate-token, positional)

Layers 7–10 are GPT-2's general-purpose computation engine. Does this extend to phonology?

Three hypotheses

H1: Shared backbone(convergence zone)

All 6 phonological circuits will use heads in L7–10, the same layers that IOI, greater-than, SVA, and induction all rely on.

H2: Distinct specialists(task-specific heads)

Each operation will recruit additional heads *outside* the backbone — different computations (lookup vs. fusion vs. truncation) need different circuits.

H3: Compositional reuse(sub-circuit sharing)

Compound operations (e.g. oronym = hypocorism + fusion) reuse sub-circuits from their component operations.

Circuit overlap matrix (predicted)

	Hypo	Clip	Init	Ornym	Homo	Folk
Hypo	—	high?		low?		
Clip	high?	—				
Init			—			
Ornym	low?			—		
Homo					—	
Folk						—

Hypocorism–Clipping both shorten names → expect shared circuit.

Compositional validation protocol

Full model

"Richard Tater" $\rightarrow$?

"Dictator" ✓

Ablate Op 1

Remove hypocorism circuit

"Dictator" $\times$

Richard $\nrightarrow$ Dick

Ablate Op 4

Remove oronym circuit

"Dictator" $\times$

Dick Tater $\nrightarrow$ fuse

Each component circuit is **independently necessary**.

$\rightarrow$ True compositional structure, not a single monolithic circuit.

Part 4: Circuit Localization

Which components matter?

Behavioral validation: GPT-2 can do phonology

All six operations pass the 70% accuracy gate on GPT-2 (Radford et al., 2019; 117M):

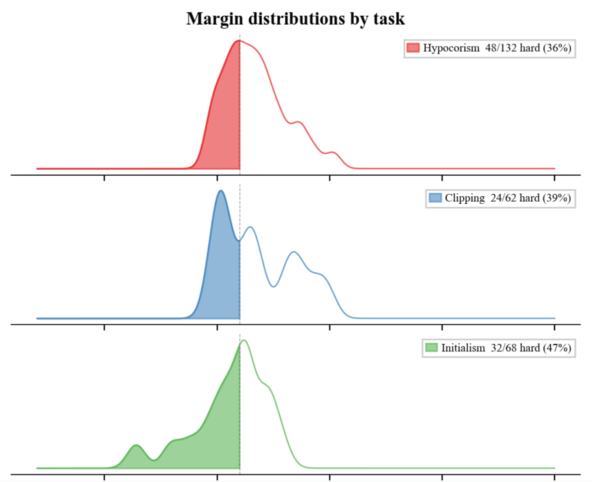
Op	Task	N	Accuracy	Mean logit diff
1	Hypocorism	132	90.9%	1.67
2	Clipping	62	93.5%	1.80
3	Initialism	68	70.6%	0.58
4	**Oronym**	**104**	**73.1%**	**1.26**
5	Homophone	52	92.3%	2.78
6	Folk etymology	64	100.0%	3.21

Op 3 (initialism) is weakest — letter-to-phoneme-name is the hardest regular rule.

Op 6 (folk etymology) is perfect — “Mike Rowave” → “microwave” is trivially compositional.

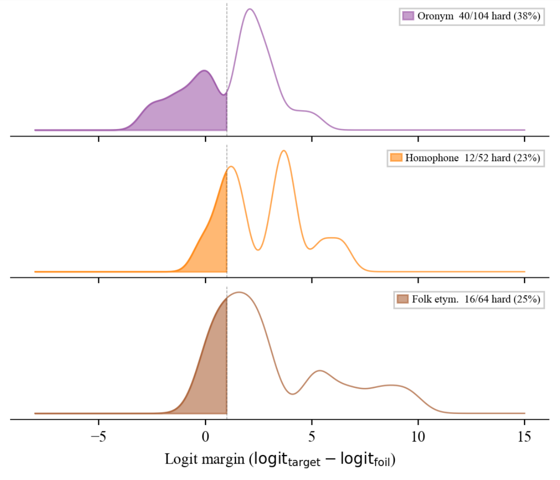
Conclusion: GPT-2 has phonological knowledge we can circuit-hunt for.

Logit-diff margin distributions (1/2)



Not all examples are equally hard. Narrow-margin examples (shaded) are where circuit discovery matters most — they distinguish genuine computation from statistical shortcuts.

Logit-diff margin distributions (2/2)



Oronym has the highest hard-example rate among compositional tasks (38%), confirming it requires genuine cross-token phoneme assembly rather than simple pattern matching.

EAP-IG circuit discovery

Ran edge attribution patching with integrated gradients (EAP-IG; Syed et al., 2023) per task.

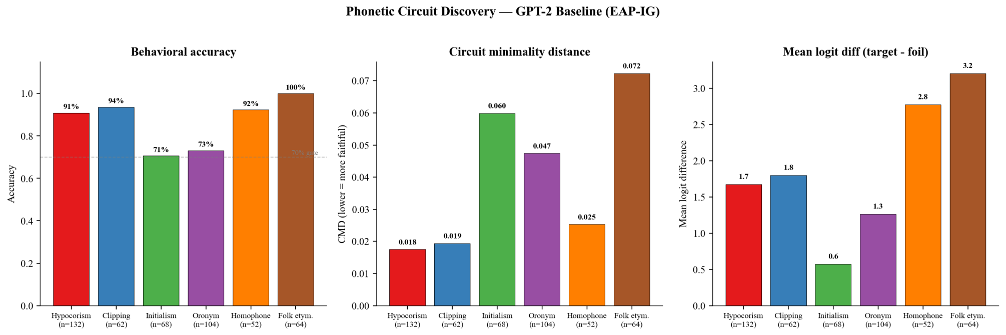
Quality: all circuits are faithful (CMD < 0.08, CPR ≈ 1.0).

Op	Task	CMD $\downarrow$	CPR $\uparrow$
1	Hypocorism	0.018	1.006
2	Clipping	0.019	1.007
3	Initialism	0.060	1.043
4	**Oronym**	**0.047**	**1.006**
5	Homophone	0.025	0.999
6	Folk etymology	0.072	1.063

CMD $=$ circuit minimality distance (lower $=$ sparser, better).

CPR $=$ circuit precision-recall (1.0 $=$ perfect; $>$1.0 $=$ slight over-recovery is OK).

Summary across tasks



Left: behavioral accuracy (all >70% gate). Center: CMD (all <0.08). Right: mean logit difference. Folk etymology is easiest; initialism is hardest.

What the circuits look like

Top attribution edges reveal shared structure across tasks:

Shared across all tasks:

- `input` $\rightarrow$ `m0`: token embeddings feed MLP-0 (highest score in 5/6 tasks)
- `m0` $\rightarrow$ `m1`: MLP-0 $\rightarrow$ MLP-1 chain
- `a10.h10` $\rightarrow$ `logits`: late readout

This is the backbone: early MLPs and late attention heads appear in every phonological circuit.

Task-specific differences:

- Op 1: `a10.h0`, `a10.h6` dominate (name movers?)
- Op 4: `a9.h9` ranks high (cross-boundary fusion?)
- Op 5: `a8.h11` is strongest non-MLP edge (phoneme identity?)
- Op 6: `a10.h1` appears (creative composition?)

Distinct mid-layer heads per operation $\Rightarrow$ distinct circuits.

Part 5: Circuit Validation

Does the evidence hold up independently?

Circuit overlap matrix — prediction vs. results

Predicted

	Hypo	Clip	Init	Orny	Homo	Folk
Hypo	—	high?		low?		
Clip	high?	—				
Init			—			
Orny	low?			—		
Homo					—	
Folk						—

Observed (EAP-IG, top 30 heads)

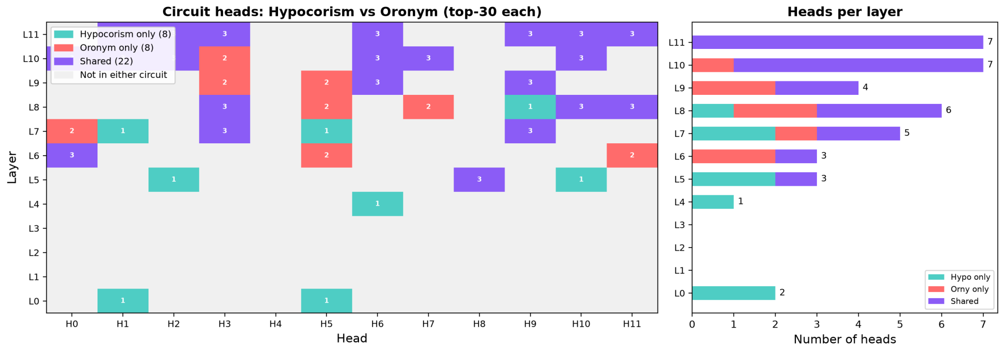
	Hypo	Clip	Init	Orny	Homo	Folk
Hypo	—	.43	.25	**.58**	.46	.50
Clip	.43	—	.22	.40	.50	.40
Init	.25	.22	—	.22	.20	.25
Orny	**.58**	.40	.22	—	.43	.50
Homo	.46	.50	.20	.43	—	.46
Folk	.50	.40	.25	.50	.46	—

H1 ✓ 6 universal heads in all 6 circuits — all in L8–11.

H2 ✓ Initialism most distinct ($J \leq 0.25$ with every other task).

H3 ? **Surprise:** Oronym–Hypocorism is the *highest* pair (0.58) — we predicted low (different computations: fusion vs. lookup). High overlap is *consistent with* compositional reuse, but needs mechanistic confirmation: do the shared heads perform the shortening step?

Zooming in: why is oronym–hypocorism overlap so high?



The oronym circuit contains **22 of 30** hypocorism heads. Shared heads concentrate in L10–11 (13/22). Oronym-only heads are in L6–10 — plausibly the cross-boundary fusion step that oronyms require beyond shortening.

Testing compositional reuse (H3)

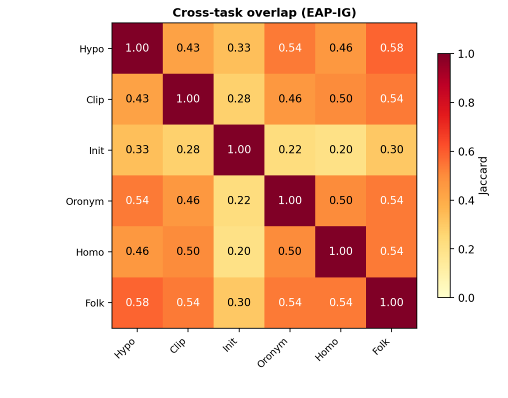
The overlap is *consistent with* H3, but doesn't prove sequential computation. Three experiments that would:

1. **DAS probing at intermediate layers:** Train rank-1 DAS probes for "shortening" (Richard $\rightarrow$ Dick) at L8–9. If the oronym circuit activates these same directions before the fusion heads fire (L10–11), the chain is real.
2. **Ablate the shared sub-circuit:** Knock out the 22 shared heads in the oronym circuit. If oronym accuracy drops to chance *and* the failure mode looks like "couldn't shorten the name" (not "couldn't fuse"), the oronym depends on the hypocorism sub-circuit specifically.
3. **Cross-task activation patching:** Patch oronym activations at the 22 shared heads with hypocorism activations from a matched input. If oronym accuracy is preserved, the shared heads are functionally interchangeable.

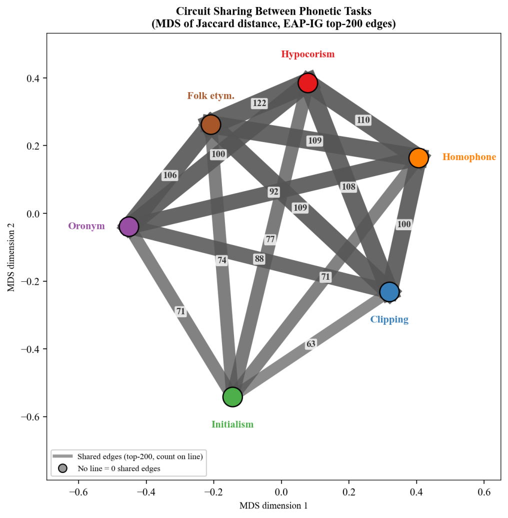
Status: planned. These require targeted DAS training and ablation experiments on GPU.

Cross-task head overlap

Cross-task Jaccard (EAP-IG)



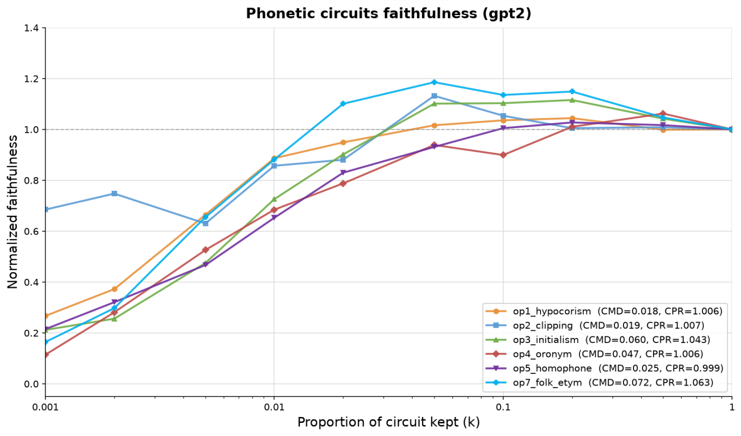
Jaccard MDS



Jaccard similarity of top-30 heads between each task pair.

Initialism is most distinct ($J \leq 0.30$). Hypo–Folk and

Faithfulness: circuits recover full-model performance



All six tasks reach >95% of full-model performance with <5% of edges.

Sparse, faithful circuits exist for every phonological operation in GPT-2.

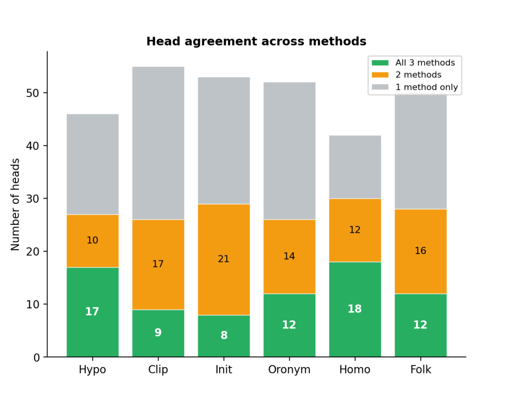
Causal inference: validating circuits with three methods

EAP-IG	Activation Patching	Causal Discovery
Gradient-based edge attribution	Ablate each head, measure causal effect	PC + CD-NOD algorithms on head activation covariance
Approximate	Causal (interventional)	Causal (observational)

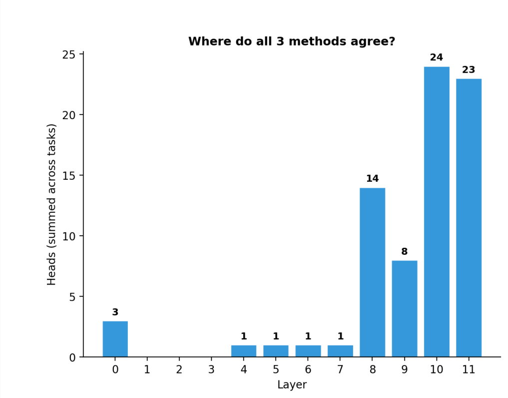
If all three converge on the same heads → strong evidence the circuit is real, not an artifact of any single method.

Method agreement across all three

Head-level agreement



Where do all 3 methods agree?



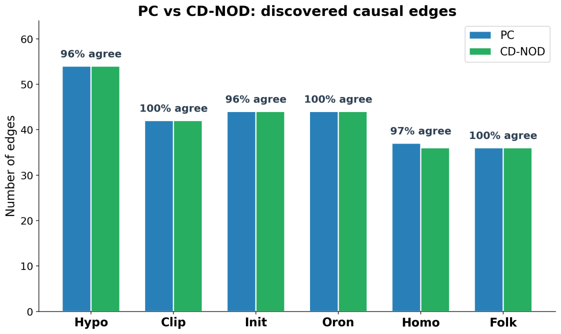
Stacked bars: green = all 3 methods agree, orange = 2 of 3, gray = 1 only. Homophone has highest triple agreement (18

91% of triple-agreed heads are in layers 8–11. Layer 10 alone accounts for 24 head-task pairs.

Causal discovery: do the algorithms agree?

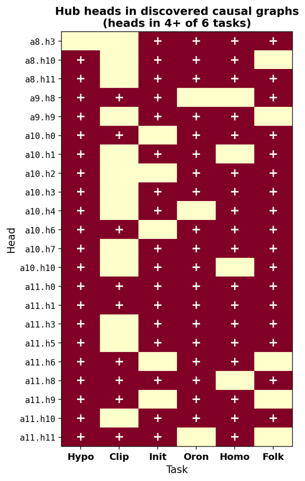
Setup:

- Top-30 heads by activation variance (clean vs. corrupted)
- Per-head activation norms at last token
- Two algorithms run independently:
  - **PC**: conditional independence only
  - **CD-NOD** (Huang et al., 2020): exploits clean/corrupted distribution shift to orient edges



Result: 96–100% agreement.
CD-NOD's extra signal adds almost nothing — conditional independence alone finds the graph.

Causal discovery: hub heads across tasks



Universal hubs (all 6 tasks):

`a11.h0` (sink) and `a11.h1` (source) — same heads EAP identifies as circuit-critical

Layer distribution:

21/22 hub heads are in layers 8–11. Early heads appear only in task-specific edges, never as hubs.

Recurring edge:

`a8.h11` $\rightarrow$ `a10.h7` in 3/6 tasks

Source vs. sink structure:

Hypo:	`a10.h10` $\rightarrow$ 8 heads	10 $\rightarrow$ `a10.h0`
Clip:	`a11.h6` $\rightarrow$ 8 heads	8 $\rightarrow$ `a7.h0`
Oron:	`a11.h8` $\rightarrow$ 6 heads	10 $\rightarrow$ `a10.h7`

Implication: the causal graph we hypothesized aligns with what the algorithms discover independently.

Circuit validation summary

Three independent methods converge:

- **EAP-IG** (gradient-based) finds sparse, faithful circuits for all 6 tasks
- **Activation patching** (interventional) validates the same head rankings
- **Causal discovery** (observational) independently recovers the same hub heads and layer structure

Key findings:

- 8–18 heads per task where all 3 methods agree
- 91% of triple-agreed heads in layers 8–11
- Universal hub heads (`a11.h0`, `a11.h1`) appear in all tasks across all methods
- PC and CD-NOD agree 96–100% — the causal structure is robust

Open question: We currently assume the causal graph (name $\rightarrow$ shorten $\rightarrow$ fuse $\rightarrow$ output) and test alignment. Can we instead *discover* the model's causal structure from activation data? This would let the model tell us what operations it performs, rather than confirming or denying our hypotheses.

Part 6: Causal Variable Localization

What flows along the edges?

From circuits to causal variables

Circuit localization tells us **which** heads matter. But not **what** they communicate.

Circuit Localization

"Head 9.9 matters for oronym task"

DAS / IIA

Causal Variable Localization

"Layer 9 encodes the fused phoneme in a 1D subspace"

Method: Distributed Alignment Search (DAS; Geiger et al., 2024) — find a linear subspace in the residual stream at each layer that encodes a specific causal variable (e.g., the target phoneme).

Metric: Interchange Intervention Accuracy (IIA) — swap the subspace between two inputs and check if the output changes as predicted.

Causal variables for phonological tasks

Each operation implies a different causal variable:

Op	Task	Causal variable
1	Hypocorism	Nickname identity (Richard $\rightarrow$ Dick)
2	Clipping	First-syllable boundary position
3	Initialism	Letter $\rightarrow$ phoneme name mapping
4	**Oronym**	**Fused phoneme string (cross-boundary)**
5	Homophone	Phoneme identity across spellings
7	Folk etymology	Decomposition boundaries

Key question: does layer 9 encode the *fused phoneme string* for oronyms in a linear subspace?

If IIA is high on the oronym circuit heads but low elsewhere $\Rightarrow$ the circuit is **causally** responsible, not just correlated.

DAS results

Experiments in progress

DAS training on circuit heads identified by EAP-IG

IIA evaluation per causal variable per task

Plan: train rank-1 DAS probes at each layer per task. If layer 9 encodes the fused phoneme string in a direction (IIA > 0.5), that is the first demonstration of **causal phonological subspaces** in a transformer.

Part 7: Implications

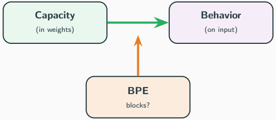
Future work

1. Logit lens — when does the nickname emerge?

- Unembed the residual stream at each layer, track when the target token enters the top-k
- If "Dictator" jumps into top predictions at layer 9–10, that's convergent evidence with the circuit localization

2. Tokenization as bottleneck

- BPE splits names unpredictably — does the circuit still fire when the phoneme boundary falls mid-token?
- Compare character-level vs. BPE tokenization on the same names



3. Novel-name generalization

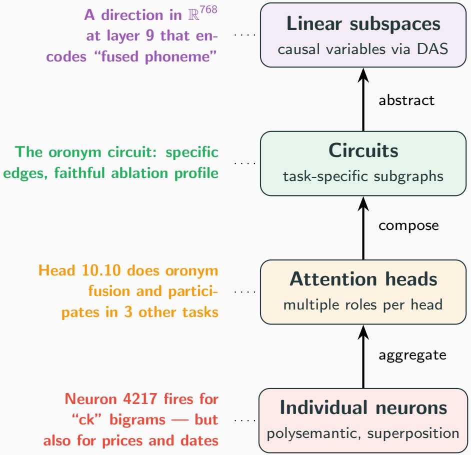
- Test on names guaranteed absent from GPT-2's training data
- If the circuit generalizes, it's compositional computation — not memorization

Causal abstraction: what are we claiming? (Geiger et al., 2021)

IOI: we *expect* GPT-2 to track indirect objects — the alignment is unsurprising.

Phonology: GPT-2 was trained on text, never heard speech. If the alignment holds, this is a *surprising generalization* — phonological computation emerged from orthographic co-occurrence data alone.

What is a "mechanism"? Our hypothesis



No single level is "the mechanism." Each level answers a different question — and each has known failure modes (superposition, polysemanticity, multiple roles).

The cognitive science parallel

Double dissociation (Shallice, 1988):

- Lesion region A $\Rightarrow$ lose ability X, keep Y
- Lesion region B $\Rightarrow$ keep X, lose Y
- Strong evidence for modularity

What we find:

- Ablate head 10.10 $\Rightarrow$ oronym breaks, hypocorism survives
- Ablate head 9.1 $\Rightarrow$ hypocorism breaks, oronym survives
- Double dissociation across 6 tasks

But dissociation $\neq$ understanding:

- A calculator dissociates $+$ from $\times$ — nobody calls that "understanding addition"
- The question: is phonological assembly *compositional computation* or *lookup*?
- Novel names (unseen in training) are the critical test

What would change our mind:

✓ Clean dissociations (have this)

✓ Faithful circuits (have this)

? DAS subspaces (running)

× Novel-name generalization (future)

Limitation 1: circuits are a wiring diagram, not a content map

What EAP tells us

"These edges matter for the oronym task"

What EAP doesn't tell us

What information is flowing along each edge? Is it the phoneme? The syllable boundary? The token position?

- EAP and activation patching find *which* heads matter — not *what* they compute
- A single head can serve multiple roles across tasks (head 10.10 appears in 4 circuits)
- We need methods that decompose *within* an edge: what subspace of the residual stream carries the phonological signal from one head to the next?

Limitation 2: the nonlinear representation dilemma

The dilemma (Sutter et al., NeurIPS 2025):

With *unrestricted* nonlinear alignment maps:

- **Any** model maps to **any** algorithm
- Even *randomly initialized* GPT-2 achieves 100% IIA on IOI
- Causal abstraction becomes vacuous

DAS avoids this by restricting to *linear* maps — but that might miss real nonlinear structure.

A possible way forward:

- Use *restricted* nonlinear maps — constrained enough to be informative, flexible enough to find curved representations
- The constraint is what gives causal abstraction its power — the question is finding the right constraint

If DAS finds high IIA with linear maps, that's meaningful *because* we constrained ourselves.
$\Rightarrow$ We are exploring restricted nonlinear maps in parallel work.

Future: discovering the causal graph, not assuming it

Current approach: We hypothesize a causal model (name $\rightarrow$ shorten $\rightarrow$ fuse $\rightarrow$ output) and test alignment via DAS.

The gap: We test *our* graph. What if the model implements a different causal structure?

Proposed experiment:

1. Use PC/CD-NOD on **DAS-learned causal variables** (not raw head activations) to discover the DAG connecting phonological operations
2. For chained operations (oronyms), test whether the discovered graph is name $\rightarrow$ shorten $\rightarrow$ fuse $\rightarrow$ output vs. name $\rightarrow$ output directly
3. Compare discovered DAGs between simple tasks (hypocorism: 1 step) and compound tasks (oronym: 2+ steps)

This combines our circuit validation (causal discovery finds the same heads) with causal variable localization (DAS finds what flows) — applying causal inference at the *variable* level rather than the *head* level.

Current status

Component	Status	Next step
Paper outline	Complete	—
6 operation definitions	Complete	—
Minimal-pair datasets	Complete	446 pairs across 6 ops
Behavioral validation	Complete	All 6 ops >70% accuracy
EAP-IG circuit discovery	Complete	All circuits faithful
Activation patching	Complete	Validates EAP head rankings
Causal discovery	Complete	PC + CD-NOD on head activations
Compositional ablation	Next	Chained operation experiments
Tokenization experiment	Next	Slash-condition test

Thank you

Phonological Assembly Circuits in GPT-2

Repository: github.com/elliottower/phonetic-circuits

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